

The identity $f^{abc} f^{def} U^{be} U^{cf} = N_c S^{ad}$

Correct, with $S^{ad} = U^{ad}$; the two-line proof, and what it does and does not license

$$U^{ab}(z) = 2 \operatorname{tr}(t^a V(z) t^b V^\dagger(z)) \text{ — the adjoint Wilson line; } U^T U = 1; \quad f^{abc} f^{gbc} = N_c \delta^{ag}$$

The identity is correct, and $S^{ad}(z)$ is nothing new — it is $U^{ad}(z)$ itself, the adjoint Wilson line, in that index order (the transpose is wrong: $\|M - N_c U^{ad}\| = 1.8 \times 10^{-15}$ versus $\|M - N_c U^{da}\| = 5.39$). It requires **both** Wilson lines at the **same** point, so invoking it is taking the $x \rightarrow z$ limit — which is exactly why it isolates the UV, and exactly why the exact remainder must be kept separately. With your index names it reads $f^{adc} f^{fbe} U^{db}(z) U^{ce}(z) = N_c U^{af}(z)$.

1. The two-line proof

The only input is that the adjoint line preserves the structure constants,

$$f^{abc} = U^{aa'}(z) U^{bb'}(z) U^{cc'}(z) f^{a'b'c'}$$

(verified to 7.8×10^{-16} ; note a generic $SO(8)$ rotation does *not* satisfy this — only a genuine adjoint $SU(3)$ line does). Multiply by U^{ad} and use $U^{ad} U^{aa'} = \delta^{da'}$:

$$f^{def} U^{be} U^{cf} = U^{gd} f^{gbc}.$$

Contract with f^{abc} and use $f^{abc} f^{gbc} = N_c \delta^{ag}$ (verified to 1.3×10^{-15}):

$$f^{abc} f^{def} U^{be}(z) U^{cf}(z) = U^{gd} f^{abc} f^{gbc} = N_c U^{ad}(z). \quad \blacksquare$$

2. It holds only at coincident points — which is the whole point

With the two lines at different points there is no collapse at all:

$$M^{ad} = f^{abc} f^{def} U^{be}(x) U^{cf}(z) : \quad \|M - N_c U(x)\| = 2.59, \quad \|M - N_c U(z)\| = 2.91, \quad \|M\| = 0.72.$$

So the identity is precisely the statement that the two lines have merged. Using it to define the UV term is legitimate; using it on the whole row is not, which is why your $2a/2b$ split is the right construction and not just a convenience.

3. Applied to your row

Match the index names: the identity's $(a, b, c) \rightarrow (a, d, c)$ of your first f , and $(d, e, f) \rightarrow (f, b, e)$ of your second. Then $U^{be} U^{cf} \rightarrow U^{db} U^{ce}$, exactly the pair in your row, and

$$f^{adc} f^{fbe} U^{db}(z) U^{ce}(z) = N_c U^{af}(z) \quad (\text{checked: } 1.3 \times 10^{-15}),$$

so your $2a$ colour structure

$$\mathcal{C}_{2a}^{af} = -f^{adc} f^{fbe} U^{db}(x) U^{ce}(z) + N_c U^{af}(x) \xrightarrow{x \rightarrow z} -N_c U^{af}(z) + N_c U^{af}(x) \rightarrow 0.$$

Consistent with the direct check: $\|\mathcal{C}_{2a}\| = 1.1 \times 10^{-15}$ at $U(z) = U(x)$.

4. A useful extra: $2a$ is a dipole of strength $\frac{N_c}{2}$

For *any* tangent variation δU of an adjoint Wilson line, the same adjointness relation gives an exact linearisation:

$$-f^{adc} f^{fbe} U^{db} \delta U^{ce} = -\frac{N_c}{2} \delta U^{af} \quad (\text{exact; max relative deviation } 4 \times 10^{-7} \text{ over 30 random directions})$$

So to leading order in the separation

$$\mathcal{C}_{2a}^{af} = \frac{N_c}{2} [U^{af}(x) - U^{af}(z)] + O((z - x)^2),$$

with strength $\frac{N_c}{2}$, **not** N_c . The deviation from this form is $O(|z - x|)$, as it should be:

$ z - x $	$\ \mathcal{C}_{2a}\ $ exact	rel. deviation from the $\frac{N_c}{2}$ form
3×10^{-1}	6.64×10^{-1}	4.2×10^{-1}
10^{-1}	2.41×10^{-1}	1.3×10^{-1}
10^{-2}	2.46×10^{-2}	1.3×10^{-2}
10^{-3}	2.46×10^{-3}	1.3×10^{-3}
10^{-4}	2.46×10^{-4}	1.3×10^{-4}

How to use the two forms. Put the exact one, $-f^{adc}f^{fbe}U^{db}(x)[U^{ce}(z) - U^{ce}(x)]$, in the note — it is an identity, and it is what you integrate. Quote the $\frac{N_c}{2}[U(x) - U(z)]$ form alongside it as the reason **2a** is UV finite: a dipole that closes linearly as $x \rightarrow z$ beats the $\frac{1}{(x-z)^2}$ of the kernel once $d^2z = |z - x|d|z - x|d\phi$ is counted, and the leftover is odd in $\widehat{z - x}$ so the angular average removes it as well. That is the $|z - x|^2$ falloff measured earlier.

5. You made the right choice of which line to keep in 2b

The identity licenses two equally exact splits:

	2b colour	2a colour	— original
(a)	$N_c[U^{af}(y) - U^{af}(x)]$	$-f^{adc}f^{fbe}U^{db}(x)[U^{ce}(z) - U^{ce}(x)]$	1.5×10^{-15}
(b)	$N_c[U^{af}(y) - U^{af}(z)]$	$-f^{adc}f^{fbe}[U^{db}(x) - U^{db}(z)]U^{ce}(z)$	8.9×10^{-16}

You took **(a)**, and that is the better one: it makes **2b**'s colour structure ***z*-independent**, so **2b** is LO-like and both of its logarithms (the UV at $x \rightarrow z$ and the infrared one at large $|z|$) come with a single clean colour factor. In **(b)** the **2b** colour still depends on z and the infrared bookkeeping is messier for no gain.

Still true, and worth repeating: the identity settles the *ultraviolet*. It does nothing for the large- $|z|$ logarithm, where $U(z) \rightarrow 1$ rather than $U(z) \rightarrow U(x)$, and there $\mathcal{C}_{2a} \rightarrow -f^{adc}f^{fbe}U^{db}(x) + N_cU^{af}(x) \neq 0$. Both **2a** and **2b** keep that logarithm, so the cancellation against $\tilde{\mathfrak{B}}_2$ has to be done on the two together. **2a** is UV finite, not finite.

Checks: `src/rowB2_identity.py` — §1 the identity and both inputs to its proof; §2 the different-points test; §3 the index matching; §4 the tangent-space linearisation and its accuracy; §5 both splits against the original. Output in `RESULTS_rowB2_identity.txt`. The UV and infrared residues themselves are in `src/rowB2_check.py` and `src/rowB2_split.py`.